Downward Löwenheim–Skolem for Countable Languages

An End-to-End Meta-Math Walkthrough

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Abstract

Close a countable seed not only under functions, but also under witnesses for every existential statement with existing parameters. Repeat countably many times so newly added parameters receive their witnesses too. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

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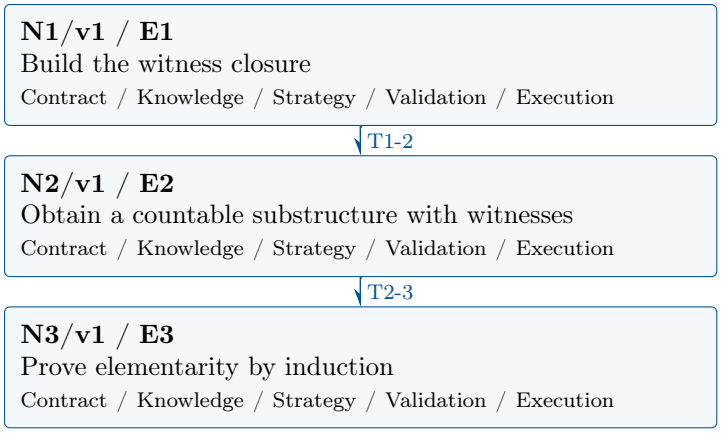
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let L be a countable first-order language with finite-arity symbols, M a nonempty L-structure, and $A \subset M$ at most countable. Prove there is an at most countable elementary substructure $N \preccurlyeq M$ containing A. Elementary means every first-order formula with parameters in N has the same truth value in N and M. If M is infinite, N can and must in this construction be countably infinite. Work in classical set theory with countable choices of witnesses.*

A. Foundations and permitted imports

Use countability of finite strings over a countable alphabet and of finite Cartesian powers of a countable set, and countable unions of countable sets. Interpret functions and relations of a substructure by restriction. The existential witness criterion for elementarity is proved here by formula induction; it is not imported as an unchecked theorem.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. There are increasing countable sets B_n such that B_{n+1} contains all function values on B_n and witnesses for all true existential formulas with parameters in B_n.

Receiving activity. N2.1

Compatibility. E1/v1 and $B = \bigcup_{n \geq 0} B_n$. Check: Closure under functions alone is insufficient for elementarity; retain the existential witness closure as a distinct export.

T2-3: N2/E2/v1 to N3. The restricted structure N on B is countable, contains A, and every existential formula true in M with parameters in N has a witness in N.

Receiving activity. N3.1

Compatibility. E2/v1 and recursive first-order semantics. Check: An existential witness true in M must be transferred using the induction hypothesis for its matrix, not simply declared true in N.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Build the witness closure

Object and role. At each stage add function values and one witness for each true existential formula over the current parameter set. **Local obligation:** There are increasing countable sets B_n such that B_{n+1} contains all function values on B_n and witnesses for all true existential formulas with parameters in B_n.

Inputs. A countable seed $B_0 \supset A$ containing one element of M, and all formulas and language symbols.

C — Contract

Publish E1/v1 only after establishing this obligation: There are increasing countable sets B_n such that B_{n+1} contains all function values on B_n and witnesses for all true existential formulas with parameters in B_n.

K — Knowledge

A countable seed $B_0 \supset A$ containing one element of M, and all formulas and language symbols. Retain the stated assumptions and import versions.

S — Strategy

At each stage add function values and one witness for each true existential formula over the current parameter set.

V — Validation

Each stage makes only countably many requests; parameters added later require new stages.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Choose one element of M and adjoin it to A to make B_0 nonempty. Given B_n, include in B_{n+1} the interpretations of all constants and of all finite-arity function symbols applied to tuples from B_n. There are countably many such requests.

N1.2 Also list every existential formula $\exists x\, \psi(x, \bar{b})$ with a finite tuple $\bar{b}$ from B_n that is true in M. There are countably many formulas and parameter tuples. Choose one witness in M for each true request and add it. With B_n itself included, the result remains countable and the construction can be repeated.

Export E1/v1. There are increasing countable sets B_n such that B_{n+1} contains all function values on B_n and witnesses for all true existential formulas with parameters in B_n.

Acceptance gate. Each stage makes only countably many requests; parameters added later require new stages.

Targeted repair. If closure is performed just once, repeat it through all natural-number stages so formulas involving newly added witnesses are covered. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Obtain a countable substructure with witnesses

Object and role. Use finiteness of parameter tuples to locate all their entries in one stage. **Local obligation:** The restricted structure N on B is countable, contains A, and every existential formula true in M with parameters in N has a witness in N.

Inputs. E1/v1 and $B = \bigcup_{n\geq 0} B_n$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: The restricted structure N on B is countable, contains A, and every existential formula true in M with parameters in N has a witness in N.

K — Knowledge

E1/v1 and $B = \bigcup_{n\geq 0} B_n$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Use finiteness of parameter tuples to locate all their entries in one stage.

V — Validation

Closure under functions alone is insufficient for elementarity; retain the existential witness closure as a distinct export.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 The union B is nonempty and at most countable. Any finite tuple from B lies entirely in some B_n, so each language function evaluated there lies in $B_{n+1} \subset B$. Constants also lie in B. Thus restriction defines an L-substructure N containing A.

N2.2 If $M \models \exists x\, \psi(x, \bar{b})$ with $\bar{b}$ in B, choose a stage containing the whole finite tuple. N1 added a witness at the next stage. Hence such a witness is in B, with the matrix formula true in M. This is the precise witness property needed next.

Export E2/v1. The restricted structure N on B is countable, contains A, and every existential formula true in M with parameters in N has a witness in N.

Acceptance gate. Closure under functions alone is insufficient for elementarity; retain the existential witness closure as a distinct export.

Targeted repair. If the proof assumes the union is closed under an operation with infinitely many arguments, restore the finite-arity language contract; the common-stage argument depends on it. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Prove elementarity by induction

Object and role. Induct through atomic formulas, Boolean operations, and existential quantification in both directions. **Local obligation:** $N \preccurlyeq M$; if M is infinite, N is countably infinite.

Inputs. E2/v1 and recursive first-order semantics. Incoming proof exports: E2/v1 at N3.1.

C **Contract**

Publish E3/v1 only after establishing this obligation: $N \preccurlyeq M$; if M is infinite, N is countably infinite.

K **Knowledge**

E2/v1 and recursive first-order semantics. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

S **Strategy**

Induct through atomic formulas, Boolean operations, and existential quantification in both directions.

V **Validation**

An existential witness true in M must be transferred using the induction hypothesis for its matrix, not simply declared true in N.

E **Execution**

Carry out activities N3.1–N3.3 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 Terms evaluate identically in N and M on tuples from N, by induction and function restriction. Atomic equalities and relations therefore agree. If formulas agree, their negations and conjunctions agree by the same truth tables.

N3.2 For an existential formula, a witness in N makes its matrix true in M by the induction hypothesis, proving the forward implication. Conversely, if the existential formula is true in M, E2 supplies a witness in N whose matrix is true in M; the induction hypothesis makes that matrix true in N. Thus the existential formula is true in N. This completes the induction for all formulas and proves elementarity.

N3.3 If M is infinite, for every positive integer r it satisfies the sentence asserting r pairwise distinct elements. Elementarity transfers each such sentence to N, so N is infinite. Since it is at most countable, it is countably infinite.

Export E3/v1. $N \preccurlyeq M$; if M is infinite, N is countably infinite.

Acceptance gate. An existential witness true in M must be transferred using the induction hypothesis for its matrix, not simply declared true in N.

Targeted repair. If elementarity is inferred from function closure, add the witness property and the two directions of the existential induction; substructure alone does not imply elementary substructure. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Build the witness closure

1. **N1.1.** Choose one element of M and adjoin it to A to make B_0 nonempty. Given B_n, include in B_{n+1} the interpretations of all constants and of all finite-arity function symbols applied to tuples from B_n. There are countably many such requests.

2. **N1.2.** Also list every existential formula $\exists x\, \psi(x, \bar{b})$ with a finite tuple $\bar{b}$ from B_n that is true in M. There are countably many formulas and parameter tuples. Choose one witness in M for each true request and add it. With B_n itself included, the result remains countable and the construction can be repeated.

B. N2 — Obtain a countable substructure with witnesses

1. **N2.1.** The union B is nonempty and at most countable. Any finite tuple from B lies entirely in some B_n, so each language function evaluated there lies in $B_{n+1} \subset B$. Constants also lie in B. Thus restriction defines an L-substructure N containing A.

2. **N2.2.** If $M \models \exists x\, \psi(x, \bar{b})$ with $\bar{b}$ in B, choose a stage containing the whole finite tuple. N1 added a witness at the next stage. Hence such a witness is in B, with the matrix formula true in M. This is the precise witness property needed next.

C. N3 — Prove elementarity by induction

1. **N3.1.** Terms evaluate identically in N and M on tuples from N, by induction and function restriction. Atomic equalities and relations therefore agree. If formulas agree, their negations and conjunctions agree by the same truth tables.

2. **N3.2.** For an existential formula, a witness in N makes its matrix true in M by the induction hypothesis, proving the forward implication. Conversely, if the existential formula is true in M, E2 supplies a witness in N whose matrix is true in M; the induction hypothesis makes that matrix true in N. Thus the existential formula is true in N. This completes the induction for all formulas and proves elementarity.

3. **N3.3.** If M is infinite, for every positive integer r it satisfies the sentence asserting r pairwise distinct elements. Elementarity transfers each such sentence to N, so N is infinite. Since it is at most countable, it is countably infinite.

□

V. WORKED ORCHESTRATION

Apply the construction to $(\mathbb{R}, +, \cdot, <, 0, 1)$ with seed $A = \mathbb{Q}$. The true formula $\exists x\,(x > 0 \wedge x^2 = 2)$ demands a witness, so the closure adds $\sqrt{2}$. Repeated stages supply witnesses for all formulas over the growing set. The resulting countable elementary substructure is more than the initial rational field; the construction does not assert that it equals a particular named subfield.

A. Boundary and hypothesis audit

The subfield $\mathbb{Q} \subset \mathbb{R}$ is a substructure of ordered fields but not an elementary substructure, because $\exists x\,(x^2 = 2)$ is false in $\mathbb{Q}$ and true in $\mathbb{R}$. Countability of the language also matters: if a distinct constant names every element of an uncountable structure, every substructure must contain all those named elements.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 conclusion: closing the seed under field operations is enough. Starting with $\mathbb{Q}$ leaves $\mathbb{Q}$, which fails the square-root existential sentence. **Diagnosis:** Knowledge recorded function closure but omitted existential witnesses. **Repair:** add one witness for every true existential request at every stage, then prove formula agreement by N3. The original theorem and seed are preserved; recheck countability and the elementarity induction.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Explore the distinction between internal and external countability in models of set theory, without interpreting the theorem as a contradiction about cardinality.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Andrew Marks, Set Theory notes for sets and ultraproducts; Berkeley, Mathematical Logic for Mathematicians, Part I for first-order semantics; Andrew Marks, Computability Theory notes for effective procedures and undecidability. The proofs below specify their own set-theoretic and semantic assumptions.